

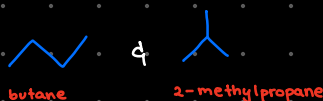
A2 Organic reactions

ISOMERS

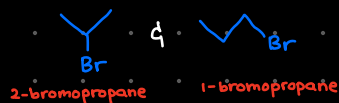
Structural

→ same molecular formula & different structural formula

① chain



② position



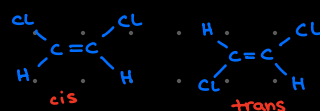
③ functional group



Stereoisomerism

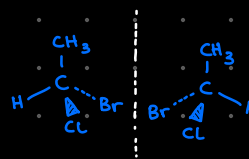
→ same structural formula & different spatial arrangement

① geometrical (E-/Z-)



due to restricted rotation

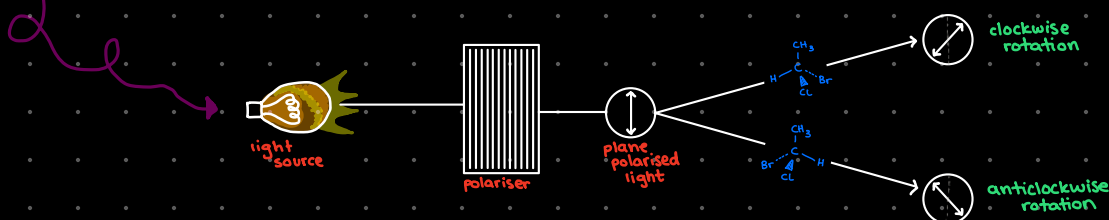
② optical



non-superimposable mirror images

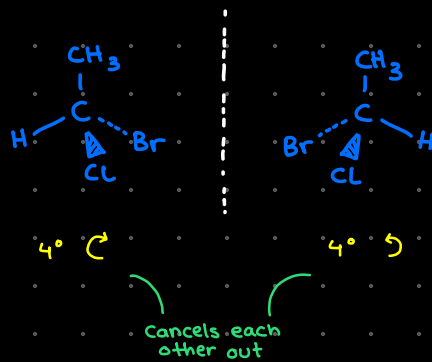
Optical isomerism:

- ↳ non-superimposable mirror images
- ↳ only in chiral compound (has a carbon attached to four different groups)
- ↳ can rotate plane polarised light



→ Racemic mixture

- a mixture containing equal amounts of both enantiomers of a chiral compound
- rotation of both enantiomers cancel each other out
- racemic mixture does not rotate plane polarised light
- it is optically inactive



why does optical isomerism matter?

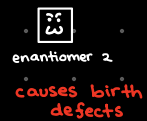
enantiomers

- rotate plane-polarised light
- can have different biological properties

1) different biological activity of enantiomers

enzymes/receptors are chiral so they can differentiate b/w enantiomers

thalidomide:



→ separation of a racemic mixture

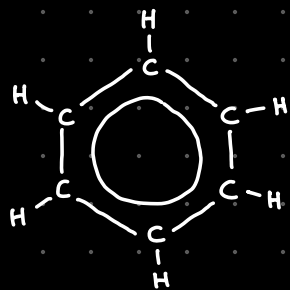
* enantiomers can have different effects; its necessary to separate a racemic mixture into its pure enantiomers

technique 1: CHIRAL RESOLUTION separating enantiomers through their difference in interactions with other chiral substances

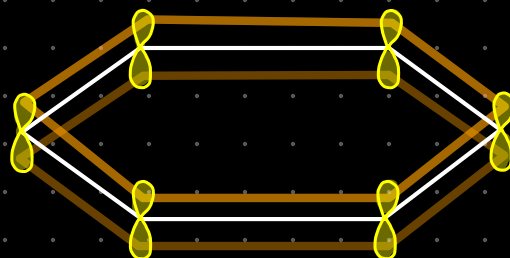
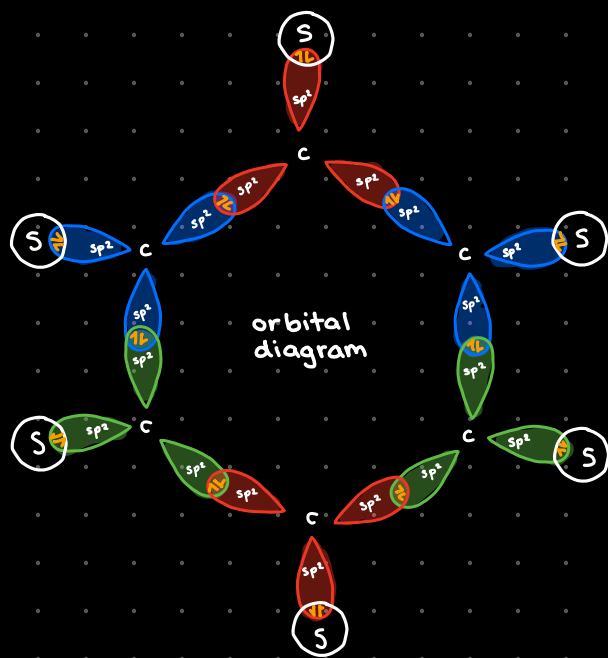
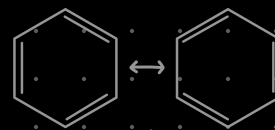
technique 2: ENANTIOMER-SPECIFIC RESOLUTION developing synthetic routes that directly only produce desired enantiomers

technique 3: CHIRAL CATALYSTS are chiral themselves & can produce a single pure optical isomer

Structure of benzene (C_6H_6)



- * regular hexagonal
- * all C-C bond lengths are the same
- * bond angle is 120°
- * each carbon is sp^2 hybridised
- * three alternate delocalised π bonds are present
- * very stable molecule due to the delocalisation of π electrons
 - ↳ thus undergoes substitution reaction easily but not addition reactions



each carbon atom in benzene ring has a unhybridised 2p orbital which forms π bond by sideways overlap

Reactions of benzene

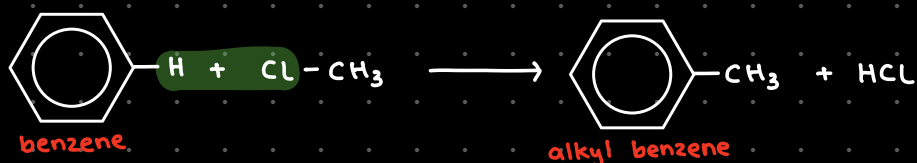
- 1) Friedel-Craft alkylation
 - 2) Friedel-Craft acylation
 - 3) Halogenoation
 - 4) Nitration
 - 5) Halogenation
- } electrophilic substitution

1) Friedel-Craft alkylation

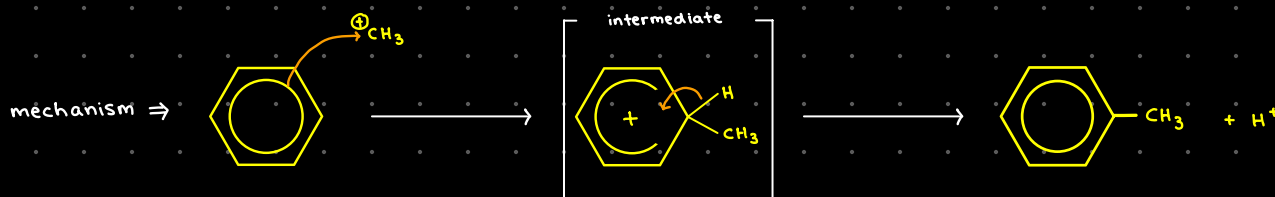
reagent: halogenoalkane

catalyst: $\text{AlCl}_3 / \text{FeCl}_3$

conditions: heat



reaction mechanism: ELECTROPHILIC SUBSTITUTION

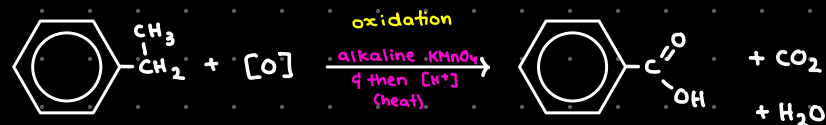


* formation of benzoic acid

① alkylation of benzene to make methylbenzene



② complete oxidation of side chain of methylbenzene

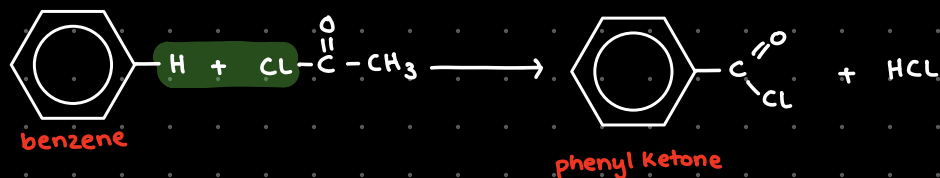


2) Friedel-Crafts acylation

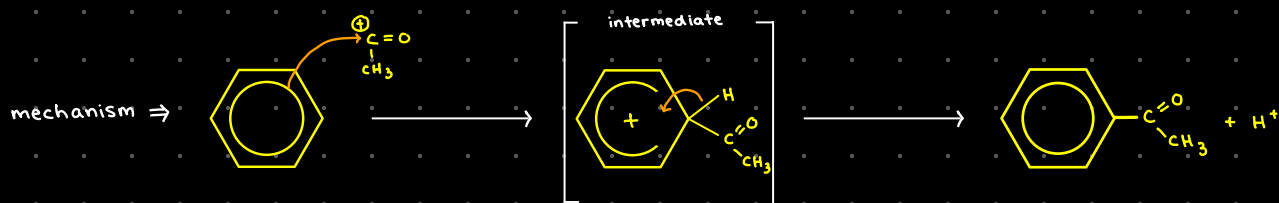
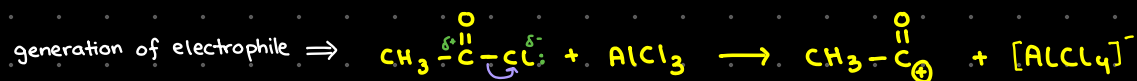
reagent: acyl chloride

catalyst: $\text{AlCl}_3 / \text{FeCl}_3$

conditions: heat



reaction mechanism: ELECTROPHILIC SUBSTITUTION

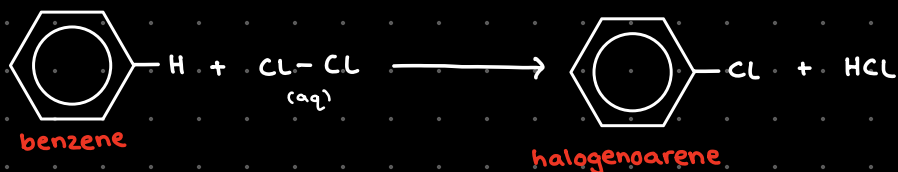


3) Halogenation

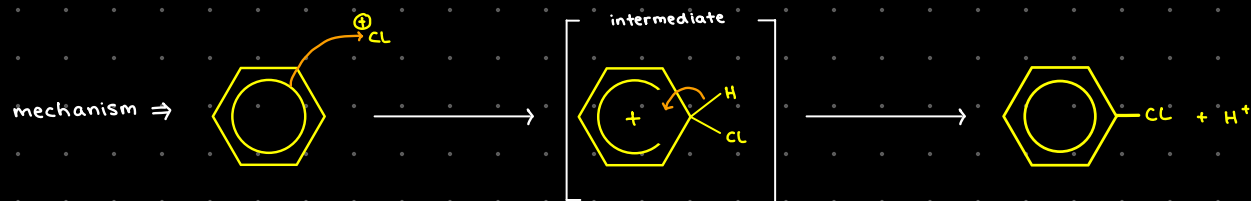
reagent: Cl_2 , Br_2

catalyst: $\text{AlCl}_3 / \text{AlBr}_3$

conditions: room temp. & anhydrous



reaction mechanism: ELECTROPHILIC SUBSTITUTION

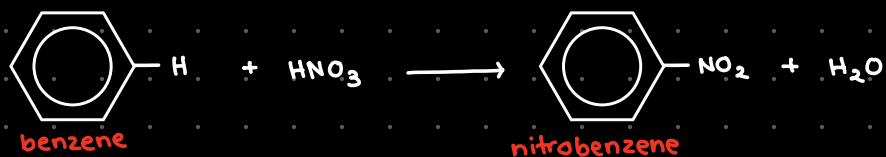


4) Nitration

reagent: conc. HNO_3

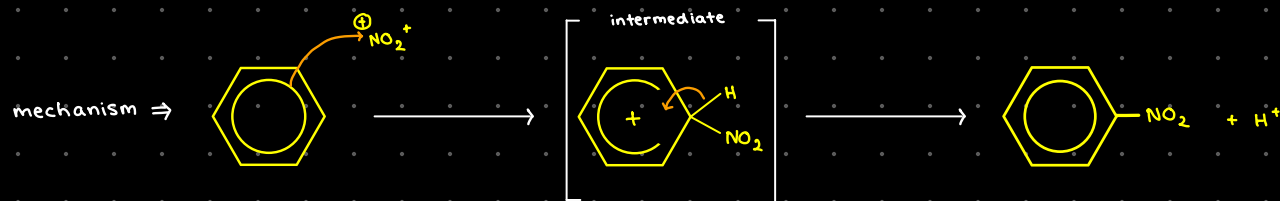
catalyst: conc. H_2SO_4

conditions: 50° under reflux



reaction mechanism: ELECTROPHILIC SUBSTITUTION

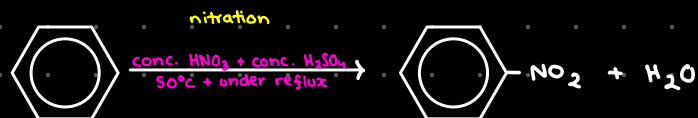
generation of electrophile $\Rightarrow \text{H}_2\text{SO}_4 + \text{HNO}_3 \longrightarrow \text{HSO}_4^- + \text{NO}_2^+ + \text{H}_2\text{O}$



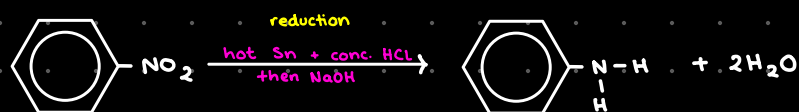
regeneration of catalyst $\Rightarrow \text{HSO}_4^- + \text{H}^+ \longrightarrow \text{H}_2\text{SO}_4$

* formation of phenylamine

① nitration of benzene to make nitrobenzene



② reduction of nitrobenzene to make phenylamine

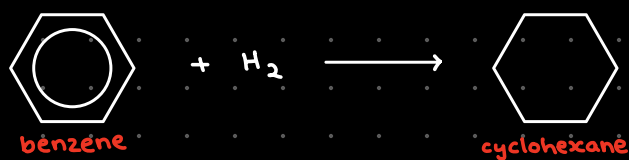


5) Hydrogenation

reagent: H_2

catalyst: Pt/Ni

conditions: heat & pressure



★ delocalised π system is broken & a saturated ring is formed

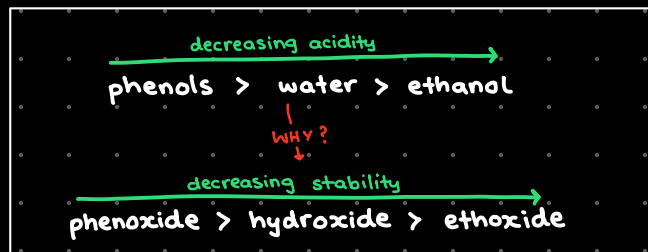
Phenol

↳ crystalline solid

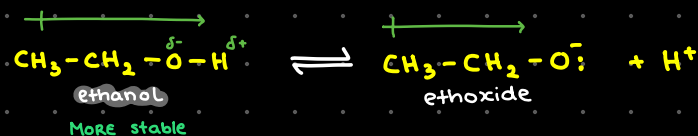
↳ weakly acidic

∴ MORE acidic than ethanol

∴ LESS acidic than metal carbonates



* oxygen atom in phenoxide ion has lone pair of e^-
* that lone pair of e^- becomes delocalised into ring
* charge is spread over the ion = MORE STABLE ion

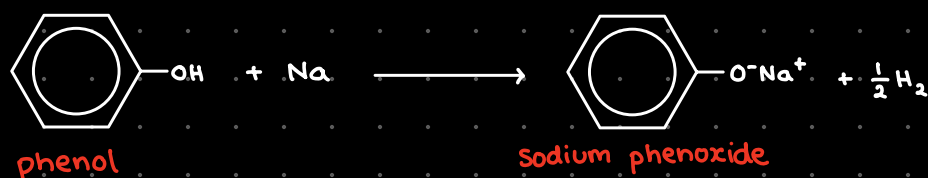


* alkyl group is electron-donating (strengthens O-H bond)
* ethanol is MORE STABLE than ethoxide

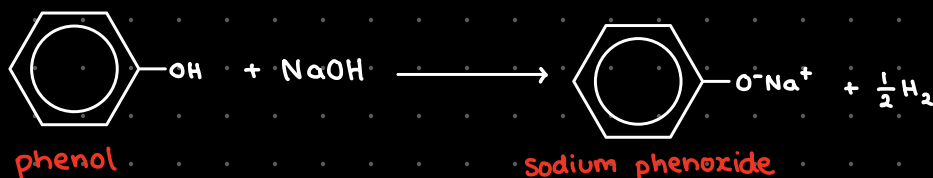
Reactions of phenols

- 1) reaction with Na
- 2) reaction with NaOH
- 3) nitration
- 4) bromination
- 5)

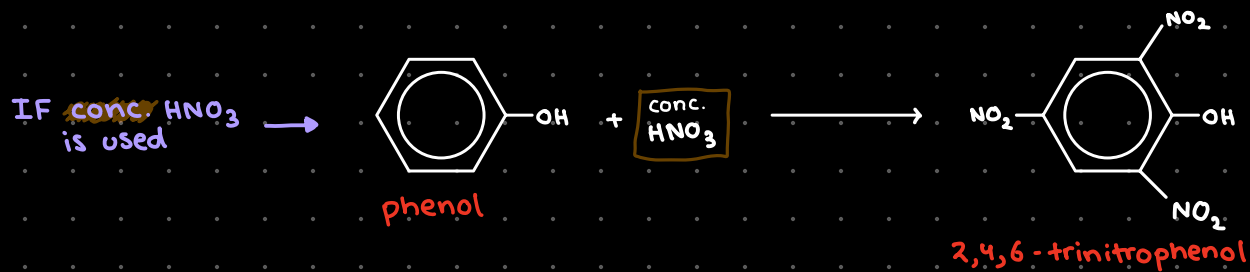
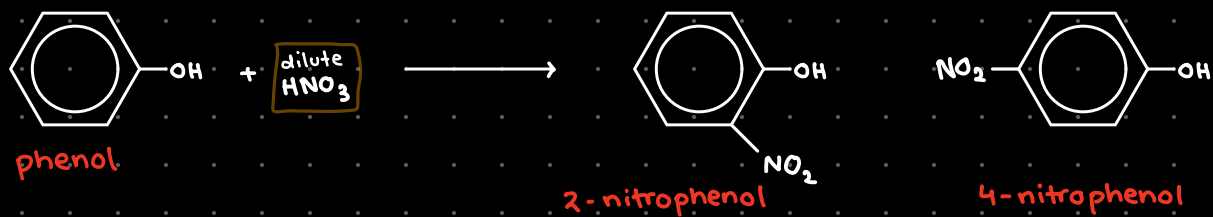
1) reaction with Na



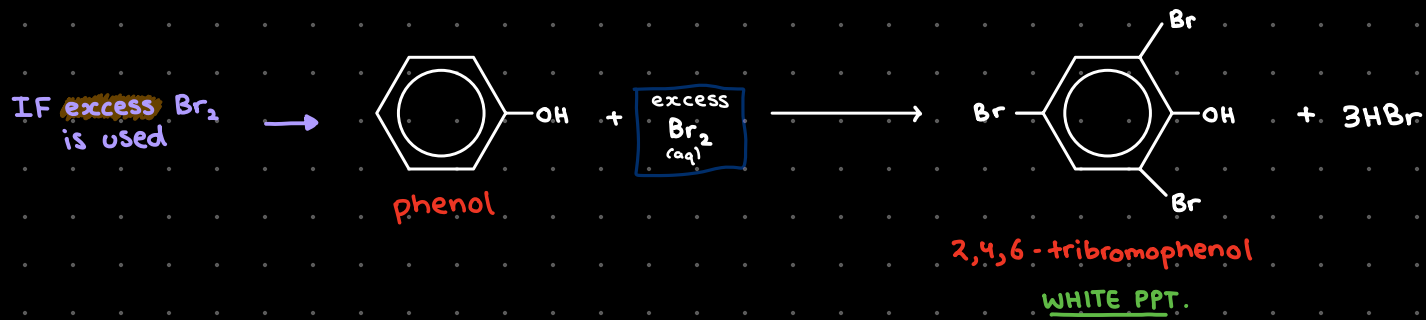
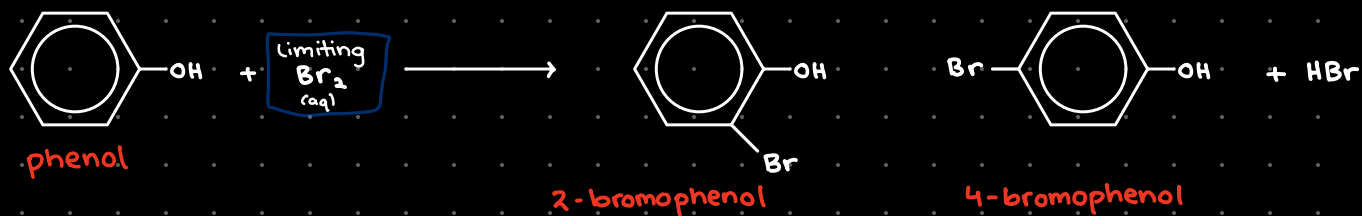
2) reaction with NaOH



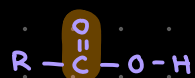
3) nitration



3) bromination

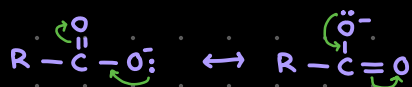


Carboxylic acid & its derivatives



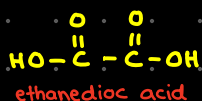
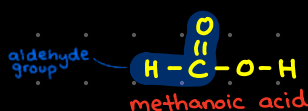
once O-H bond
is broken

C=O is electron-withdrawing so it weakens O-H bond



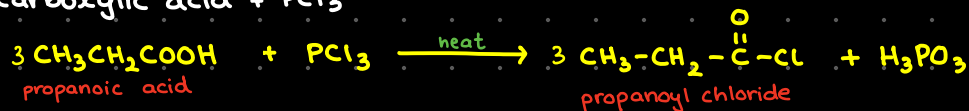
carboxylate ion is stabilised by movement of lone pair in COO (resonance)

Oxidation of methanoic acid & ethanedioic acid

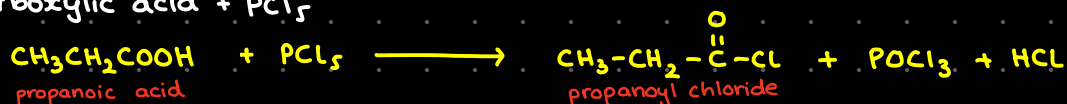


Formation of acyl chloride

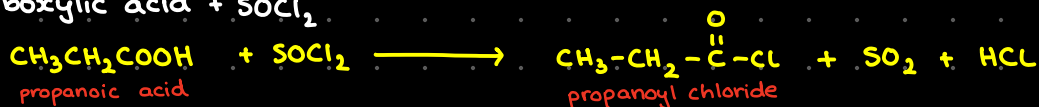
(a) carboxylic acid + PCl_3



(b) carboxylic acid + PCl_5

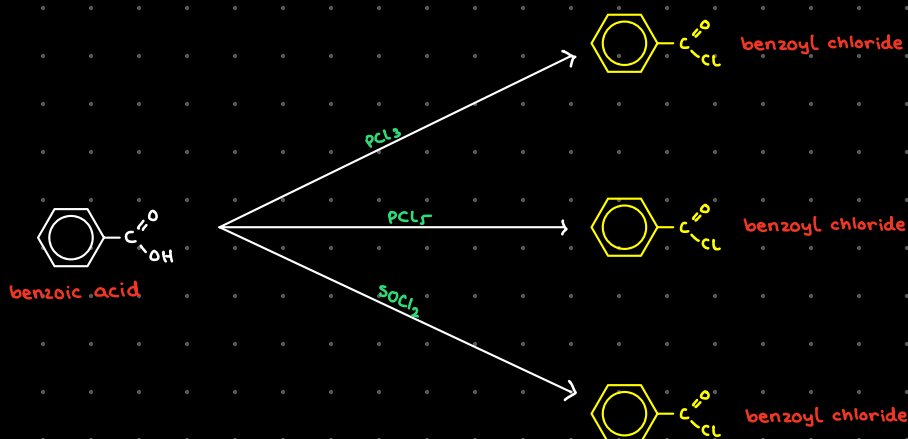


(c) carboxylic acid + SOCl_2



★ only difference between the three reactions is the by-products

Formation of benzoyl chloride

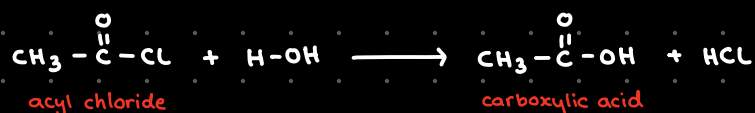


★ only difference between the three reactions is the by-products

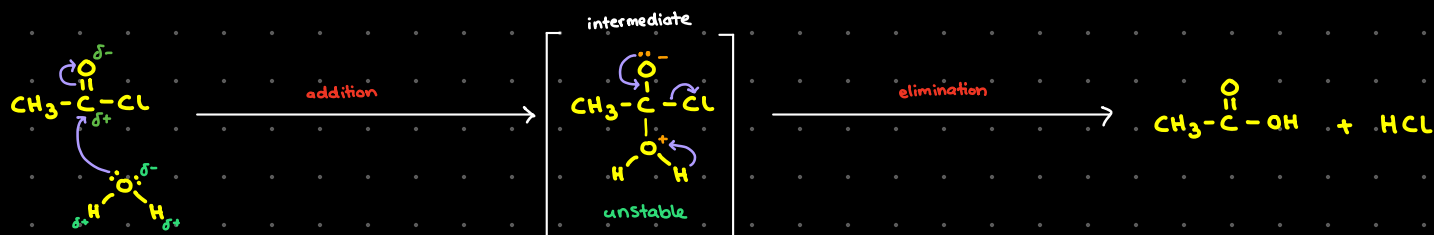
Reactions of acyl chlorides

- (1) hydrolysis
- (2) esterification

(1) hydrolysis



reaction mechanism: **ADDITION-ELIMINATION**



1) lone pair on oxygen attacks the carbonyl carbon

2) the C=O bond breaks & e⁻ pair moves to oxygen, making it negative

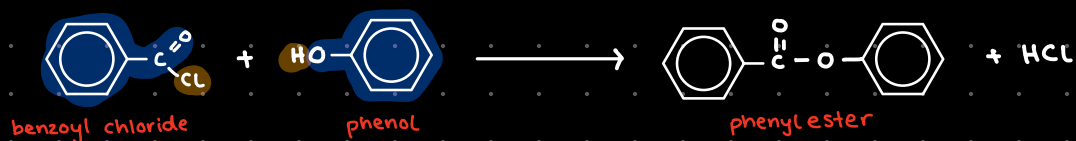
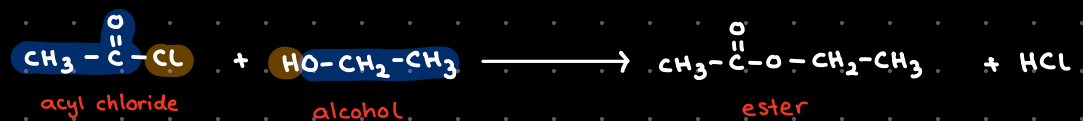
3) temporary unstable intermediate forms where carbonyl oxygen has -ve charge & water oxygen has +ve charge

4) O⁻ lone pair drops back, kicking out the Cl⁻ (leaving group)

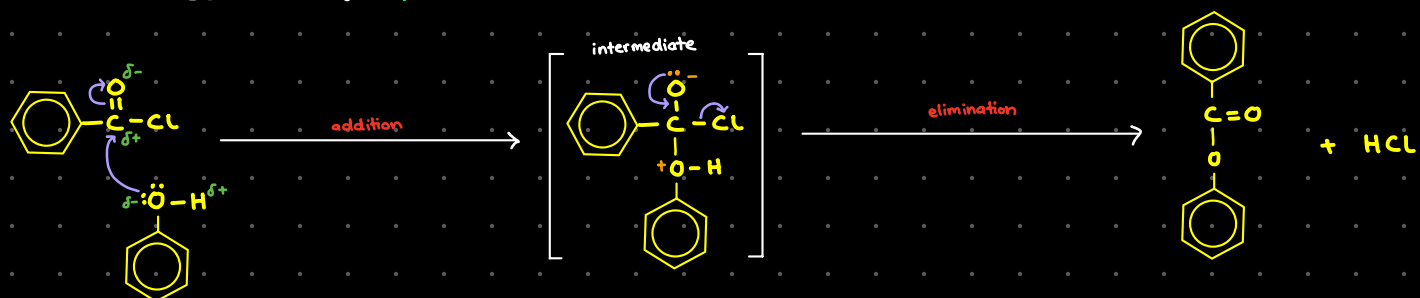
5) proton is removed from water oxygen

6) ethanoic acid is formed!

(a) esterification



reaction mechanism: **ADDITION-ELIMINATION**



1) lone pair on oxygen attacks the carbonyl carbon

2) the $\text{C}=\text{O}$ bond breaks & e^- pair moves to oxygen, making it negative

3) temporary unstable intermediate forms where carbonyl oxygen has -ve charge & phenol oxygen has +ve charge

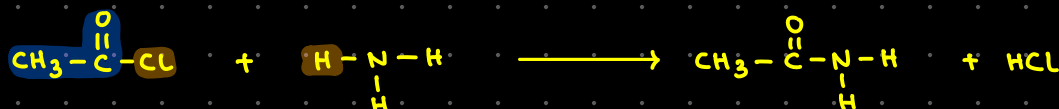
4) O^- lone pair drops back, kicking out the Cl^- leaving group

5) proton is removed from phenol oxygen

6) phenyl ester is formed!

Preparation of amides

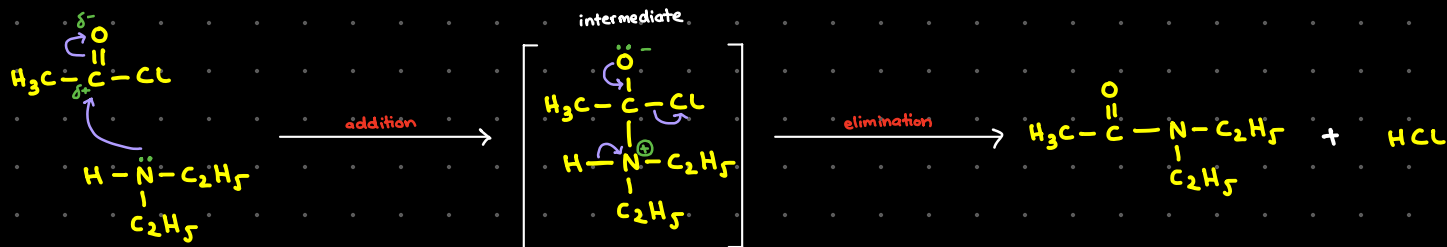
(a) acyl chloride + NH_3



(a) acyl chloride + amide



reaction mechanism: **ADDITION-ELIMINATION**



1) lone pair on nitrogen attacks the carbonyl carbon

2) the $\text{C}=\text{O}$ π bond breaks & electron pair moves to oxygen, making it $-\text{ve}$

3) temporary unstable intermediate forms where oxygen has a $-\text{ve}$ charge & nitrogen has a $+\text{ve}$ charge

4) O^- lone pair drops back, kicking out the Cl^- (leaving group)

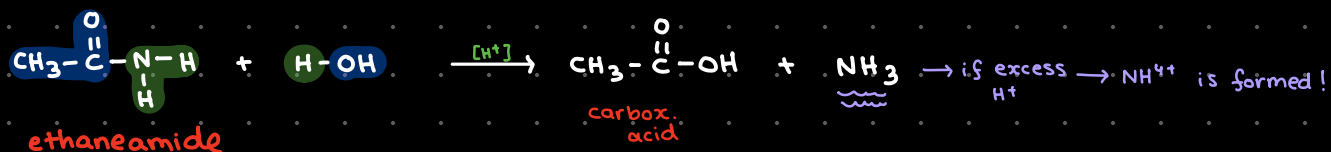
5) proton is removed from nitrogen

6) N,N-diethylamide is formed!

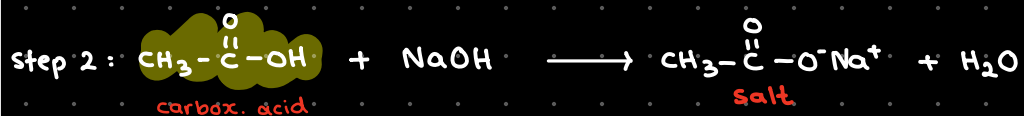
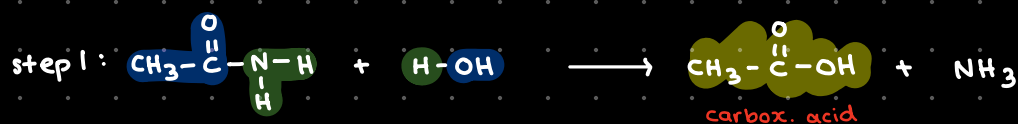
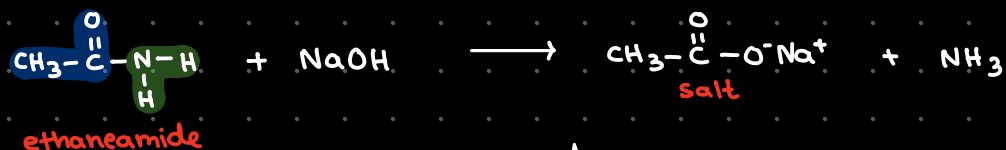
Reactions of amides

- (1) acidic hydrolysis
- (2) basic hydrolysis
- (3) reduction

(1) acidic hydrolysis



(2) basic hydrolysis

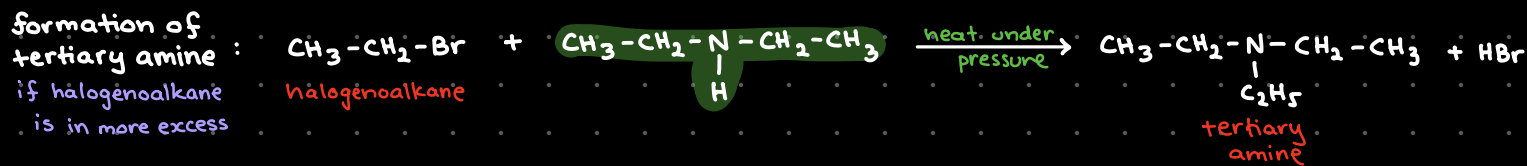
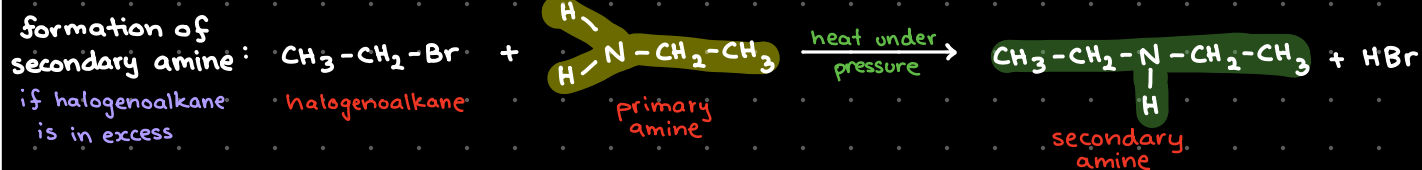
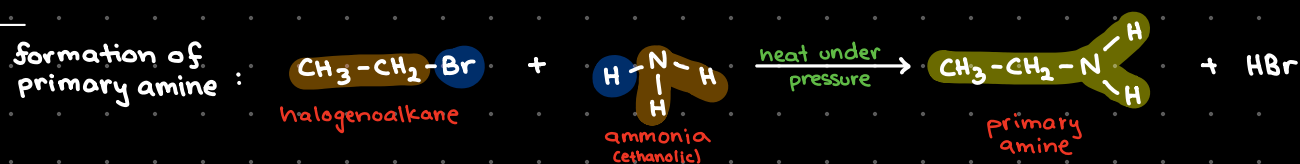


Nitrogen compounds

Formation of amines

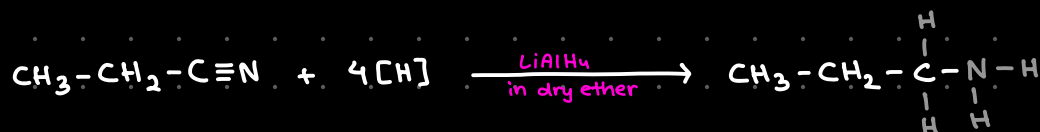
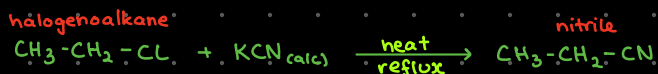
- (1) halogenoalkane + ammonia
- (2) reduction of nitriles
- (3) reduction of amides

(1) halogenoalkane + ammonia

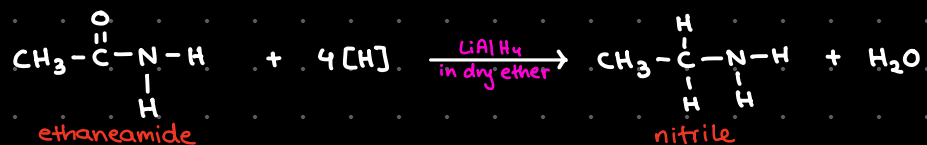


(2) reduction of nitriles

How will we prepare the nitrile?



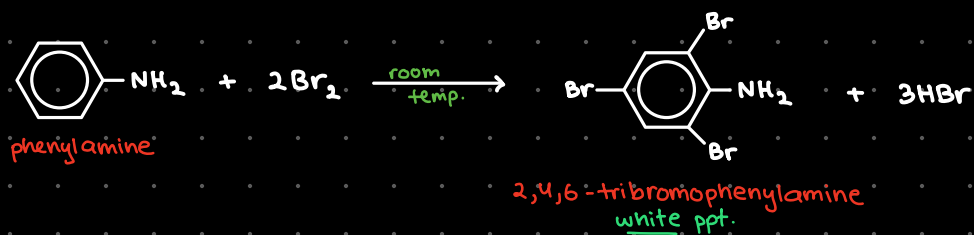
(3) reduction of amides



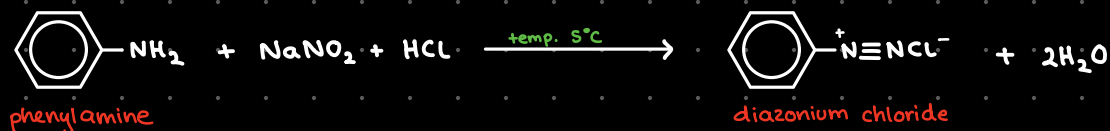
Reactions of phenylamine

- (1) bromination
- (2) diazotisation

(1) bromination



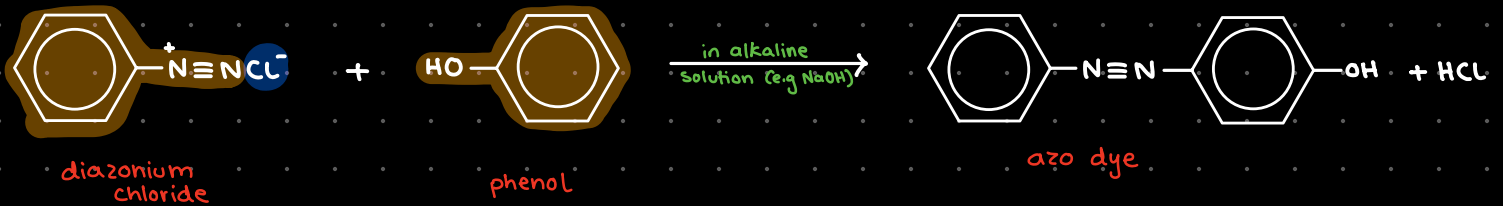
(2) diazotisation



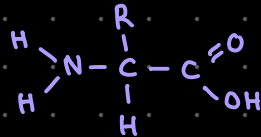
when temp.
goes $>10^\circ\text{C}$



Formation of dyes



Amino acids

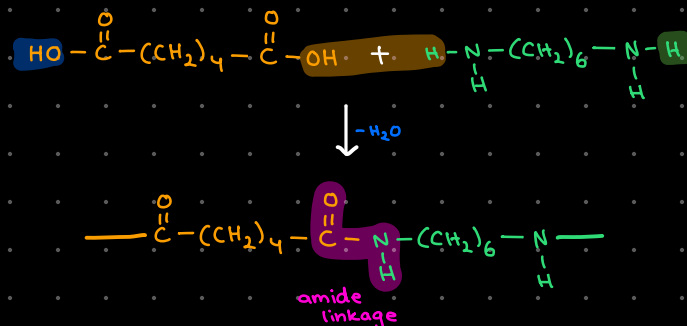
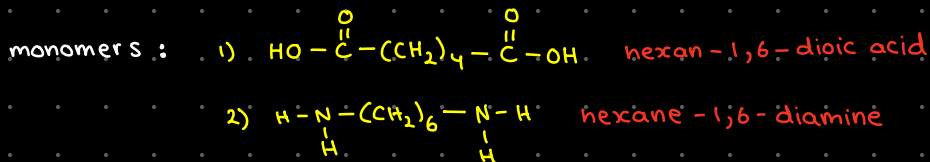


→ condensation polymerisation

- ↳ monomers: amino acids
- ↳ forms polypeptides/proteins
- ↳ amide linkage
- ↳ biodegradable polymers (can be hydrolysed)

- * dipeptide: made of 2 amino acids
- * polypeptide: made of many amino acids

e.g. Nylon - 6,6

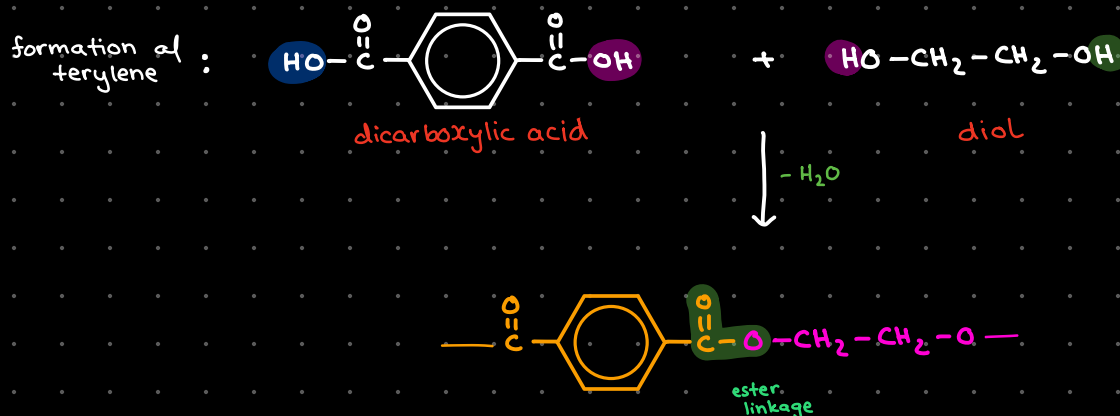


i'm bored of polymerisation so i can't be bothered to write more

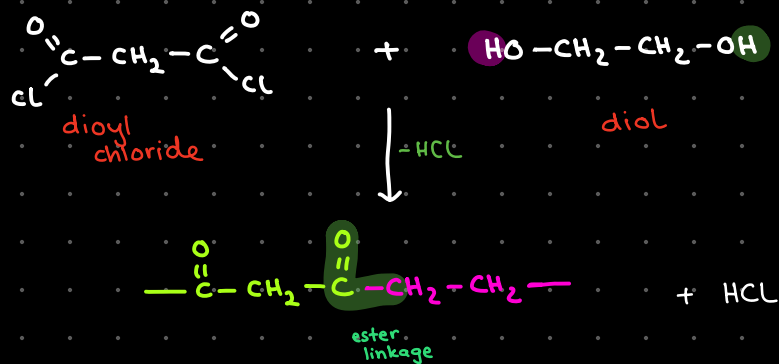
Formation of polyesters

- (1) diol + dicarboxylic acid
- (2) diol + diacyl chloride
- (3) hydroxycarboxylic acid

(1) diol + dicarboxylic acid



(2) diol + diacyl chloride



(3) hydroxycarboxylic acid

